ROP PREDICTION

Real-Time Drilling Optimization with ML

DATABRICKS SOLUTION ACCELERATOR

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ML-POWERED RATE OF PENETRATION PREDICTION & REAL-TIME DRILLING OPTIMIZATION

ROP Prediction delivers an end-to-end ML pipeline — from MSEEL field data ingestion through XGBoost training, Spark Structured Streaming inference, and an operational dashboard — to optimize drilling and reduce well costs by up to $500K+ per well.

◆ BUSINESS VALUE

- **Save $500K+ per well** by optimizing rate of penetration through real-time ML predictions
- **Detect drilling hazards early** — high MSE, stuck pipe risk, and inefficient drilling flagged automatically
- **Track model performance** with MLflow registry showing version history, RMSE, and R^2 metrics
- **Integrate with SAP ERP** for work orders, equipment BOM, procurement tracking, and vendor contracts
- **Enable real-time scoring** with Spark Structured Streaming inference on live drilling telemetry

◆ CHALLENGES ADDRESSED

- **Suboptimal drilling parameters** — no real-time guidance on weight-on-bit, RPM, and flow rate
- **Delayed hazard detection** — problems identified only after costly stuck pipe or bit damage
- **No ML in the loop** — drilling decisions based on experience alone without predictive models
- **Disconnected data** — MSEEL field data, model predictions, and work orders in separate systems
- **Manual performance tracking** — no automated comparison of actual vs. predicted ROP

◆ WHO IT'S FOR

Drilling Engineer
Real-time ROP predictions, MSE monitoring, and parameter optimization

Data Scientist
End-to-end ML pipeline with XGBoost, feature engineering, and MLflow tracking

Drilling Superintendent
Fleet-level KPIs, hazard alerts, and drilling efficiency across wells

Rig Manager
SAP ERP integration for work orders, equipment BOM, and procurement

2	10	7	4
MSEEL Wells	Drilling Parameters	Dashboard Tabs	Model Versions

◆ QUICK START GUIDE

1. **Clone Repository** — Clone the ROP Prediction repo to your local machine
2. **Run Infrastructure SQL** — Create Delta tables and Lakebase PostgreSQL schema
3. **Deploy the App** — Run databricks apps deploy to launch the dashboard
4. **Run ML Pipeline** — Ingest, train, and start the streaming scorer

◆ GETTING THE MOST OUT OF ROP PREDICTION

Monitor Drilling Command
Track real-time ROP, MSE, WOB, and drilling efficiency with hazard detection

Analyze Deep Analytics
Compare actual vs. predicted ROP and track MLflow model versions

Ask Drilling Agent
Chat with the Claude-powered assistant for drilling optimization

Review Live Alerts
Monitor streaming hazard feed with severity-based filtering

◆ TARGET INDUSTRY

Upstream Oil & Gas — Drilling operations in unconventional shale plays. Built with real MSEEL (Marcellus Shale Energy and Environment Laboratory) field data from West Virginia.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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